MOHAMED AMINE HERCH

Industrial Electronics Engineer | Mobile Robotics & Embedded Systems

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PROFILE

Industrial Electronics Engineer specializing in mobile robotics, industrial automation, and embedded systems. Hands-on experience integrating autonomous mobile robots and developing robotic systems using ROS 2 and Open-RMF, with real-world experience in navigation, localization, robot communication, and system troubleshooting. Skilled in C/C++, Python, industrial communication, sensors and actuators, and real-time robotic systems.

TECHNICAL SKILLS

Robotics & Fleet Management: ROS 2 Jazzy, Nav2, SLAM, Open-RMF, AMR/AGV Integration, Autonomous Navigation
Digital Twin & 3D: Unity 6.3 LTS, Real-Time 3D Visualization, Digital Twin, ROS-TCP-Connector, 3D Scanning
Mechatronics & Automation: Sensors & Actuators, Electromechanical Systems, Motors & Motor Control, Industrial Automation, Measurement
Computer Vision: YOLO, ByteTrack, Multi-Object Tracking, OpenCV, Camera–LiDAR Fusion
Embedded Systems & IoT: STM32, ESP32, ESP8266, Arduino, FPGA/VHDL, RFID, MQTT, Serial Communication
Programming & Tools: C/C++, Python (rclpy), MATLAB, Node-RED, Git, Linux, Proteus, Altium, Xilinx ISE
Prototyping & Fabrication: 3D Printing, Laser Engraving

PROFESSIONAL EXPERIENCE

Final-Year Engineering Project, CNIMI Laboratory (UQTR) | Drummondville, Canada | 2026
Heterogeneous AMR Fleet Integration and Digital Twin Supervision

- Developed and configured a heterogeneous AMR fleet management system using ROS 2 Jazzy and Open-RMF, enabling multi-robot coordination across different manufacturers.
- Developed a Unity-based Digital Twin for real-time 3D visualization, robot state monitoring, and fleet supervision through ROS-TCP-Connector.
- Integrated MiR250, MiR600, and Omron LD-90 robots into the Open-RMF framework, including maps, navigation routes, traffic rules, and fleet coordination.

Keywords: Open-RMF, ROS 2 Jazzy, Digital Twin, Unity, AMR, Multi-Robot Coordination, ROS-TCP-Connector, ARCL, Python, C++, Linux

Engineering Intern, Rosenberger | Sousse, Tunisia | 2025

- Developed autonomous navigation and mapping simulations using ROS and SLAM algorithms.
- Designed and built a physical robotic prototype integrating a Raspberry Pi, LiDAR, and DC motors with encoders.

Keywords: Raspberry Pi, ROS, LiDAR, SLAM, DC Motors with Encoders, Autonomous Navigation, Robotics

Senior Maker and Trainer, FabLab ENISo | Sousse, Tunisia | 2024 – Present

- Mentored and supervised robotics and mechatronics projects from concept and design through prototyping, integration, and testing.
- Delivered technical training sessions and hands-on workshops in robotics, electronics, and digital fabrication.
- Hands-on experience with 3D printing, laser engraving, and 3D scanning.

Intern, Leoni | Sousse, Tunisia | 2024

- Observed all stages of wiring harness production; studied assembly processes and quality control procedures.

ACADEMIC PROJECTS

Person-Following Robot: TurtleBot3, ROS 2, YOLO and Multi-Object Tracking **2026 – Present**

- Built a ROS 2 Jazzy pipeline where a TurtleBot3 detects people (YOLO26n), tracks them (ByteTrack) and follows the user-selected person at 1 m, with camera–LiDAR distance fusion, a P controller, a watchdog and an emergency stop.
- Wrote an identity layer that restores a hidden person's ID by clothing colour, plus a lost-target search mode; 57 unit tests.

Keywords: ROS 2, Gazebo, YOLO, ByteTrack, OpenCV, Visual Servoing, Python | github.com/amineherch-robotics/person_tracking

SmartTime IoT: Connected Employee Attendance System

- Developed an ESP8266-based RFID attendance system with MQTT/HiveMQ communication and Node-RED real-time monitoring.

Sign Language Recognition

- Developed a MATLAB gesture recognition system using image processing, computer vision and machine learning for classification.

Line-Following Robot

- Developed an autonomous line-following robot using Pololu encoders, sensor fusion, and PID control for accurate trajectory tracking.

EDUCATION

Engineering Degree in Industrial Electronics, National Engineering School of Sousse (ENISo) | Tunisia | 2023 – 2026

Scientific Preparatory Cycle, Preparatory Institutes for Engineering Studies of Monastir (2022 – 2023) and Sfax (2021 – 2022) | Tunisia

LEADERSHIP & ACTIVITIES

President, ENISO Team | Sousse, Tunisia | 2024 – 2025

- Led a multidisciplinary team of engineering students.
- Organized technical training sessions covering SolidWorks, MATLAB, STM32, Arduino, and ESP32.
- Coordinated major student events, including ESC V9.

LANGUAGES

Arabic: Native | French: Professional | English: Professional